

Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTrainingData/NewNewDynamic · 720 trials
Total training paths	1 · 720 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/NewNewDynamic
Report output dir	/home/jestin/ThesisRepo/ML/NewReports/NewDynamicReports/RF_Models

Sensor selection

Hands	left
Segments	thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	20 channels
Classes	6: Double_ClosedHand_Flip_OpenHand, Double_FingerTips, Double_Okay_Ring_Middle_Pinky, Double_OpenHand_Flip_ClosedHand, Double_OpenHand_LPunch_R_Punch, Double_Snap

Sensor grid (✓ enabled, ✗ disabled, — not present)

Hand / Segment	IMU_mid	IMU_prox	Flex_MCP	Flex_PIP
left palm	✗	—	—	—
left thumb	✗	✗	✓	✓
left index	✗	✗	✓	✓
left middle	✗	✗	✓	✓
left ring	✗	✗	✓	✓
left pinky	✗	✗	✓	✓
left wrist	✗	—	—	—
right palm	✗	—	—	—
right thumb	✗	✗	✓	✓
right index	✗	✗	✓	✓
right middle	✗	✗	✓	✓
right ring	✗	✗	✓	✓
right pinky	✗	✗	✓	✓
right wrist	✗	—	—	—

Enabled: 20 · Disabled: 24 (channels per enabled IMU depend on modality flags above).

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats+fft → feature dim 1080
Test size · seed · CV folds	0.2 · random_state=42 · 5-fold stratified
Sample counts	train (post-aug)=2880 · test=144

Augmentation (training only)

Enabled	yes
Copies per train trial	4
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.9, 1.1)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

CV = 5-fold cross-validation. Test = hold-out accuracy. M-* = macro-averaged, W-* = support-weighted. Prec / Rec / F1 on hold-out test.

Algorithm	CV acc	CV std	CV F1	CV F1 σ	Test	M-Prec	M-Rec	M-F1	W-Pre c	W-Rec	W-F1
Random Forest	1.0000	0.0000	1.0000	0.0000	0.9792	0.9805	0.9792	0.9791	0.9805	0.9792	0.9791

Per-class metrics (hold-out test)

=== Random Forest ===

	precision	recall	f1-score	support
Double_ClosedHand_Flip_OpenHand	1.00	0.92	0.96	24
Double_FingerTips	0.96	1.00	0.98	24
Double_Okay_Ring_Middle_Pinky	1.00	0.96	0.98	24
Double_OpenHand_Flip_ClosedHand	0.92	1.00	0.96	24
Double_OpenHand_LPunch_R_Punch	1.00	1.00	1.00	24
Double_Snap	1.00	1.00	1.00	24
accuracy			0.98	144

macro avg	0.98	0.98	0.98	144
weighted avg	0.98	0.98	0.98	144

External test (NewTestData)

Algorithm	External test acc
Random Forest	0.7583

Per-class external accuracy

Class	Random Forest	Support
Double_ClosedHand_Flip_OpenHand	0.550	20
Double_FingerTips	0.500	20
Double_Okay_Ring_Middle_Pinky	0.700	20
Double_OpenHand_Flip_ClosedHand	0.950	20
Double_OpenHand_LPunch_R_Punch	0.900	20
Double_Snap	0.950	20